1. An (n, M, d) code $\mathcal{C}$ over a field $\mathbb{F}_q$ is said to be a constant weight code is all the codewords of $\mathcal{C}$ have the same Hamming weight w. Let $\mathcal{C}$ be a constant weight code with parameters $(n, M, d = 2t+1)$ and $w = 2t+1$ over $\mathbb{F}_q$. Show the following:

$$M \leq \frac{\binom{n}{t+1}(q-1)^{t+1}}{\binom{2t+1}{t}}$$

(6 marks)

2. Consider an $[n, k, d]$ linear code $\mathcal{C}$, over $\mathbb{F}_2$, with $d = 2t+1$, for some $t \in \mathbb{N}$. Suppose that in the standard array corresponding to $\mathcal{C}$, there are no coset leaders of weight strictly larger than t.
   (a) What, then, is the probability of error under standard array decoding, when $\mathcal{C}$ is used over a BSC with cross-over probability p? (4 marks)
   (b) If standard array decoding were used with the $[7,4,3]$ binary Hamming code over a BSC with cross-over probability p, using the result in the previous section, calculate the probability of error. (2 marks)

3. Let $G = [I|A]$ be a systematic generator matrix of a linear (n, k, d) code $\mathcal{C}$ over $\mathbb{F}_q$. Show that $\mathcal{C}$ is maximum distance separable (MDS) code if and only if every square submatrix of A is non-singular (invertible). (5 marks)

4. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be (n, k_1) and (n, k_2) linear codes with generator matrices G_1 and G_2 respectively. Consider the code $\mathcal{C}$ constructed as follows:

$$\mathcal{C} = \{(\underline{u}, \underline{u} + \underline{v}) : \forall \underline{u} \in \mathcal{C}_1, \underline{v} \in \mathcal{C}_2\}.$$

   (a) Show that $\mathcal{C}$ has the following generator matrix:

$$G = \begin{bmatrix} G_1 & G_1 \\ 0_{k_2 \times n} & G_2 \end{bmatrix}$$

(4 marks)

   (b) Show that the minimum distance of $\mathcal{C}$ is $\min(2d_1, d_2)$. (2 marks)

5. Compute the gcd of $a(x) = x^4 + x + 1$ and $b(x) = x^2 + 1$, over $\mathbb{F}_2$. Hence, find polynomials, $r(x)$ and $s(x)$, such that $\gcd(a(x), b(x)) = r(x)a(x) + s(x)b(x)$ using the division algorithm. (5 marks)

6. Construct a finite field $\mathbb{F}_9$ using $\mathbb{F}_3[x]$ and an irreducible polynomial of degree 2 over $\mathbb{F}_3$. Prove that it satisfies all the properties of the finite field. (7 marks)